



2021 Caringbah EXT2 AT3 Exam

Maths extension 2 (Sydney Technical High School)



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Caringbah High School

Year 12 2021

Mathematics Extension 2

HSC Course

Assessment Task 3

General Instructions

- Reading time – 5 minutes
- Working time – 70 minutes
- Write using a black pen
- Board-approved calculators may be used
- A reference sheet is provided.
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 45

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 40 marks

Attempt Questions 6-9
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.^[SL1]

Name: _____

Class: _____

Marker's Use Only						
Section I	Section II				Total	
Q 1-5	Q6	Q7	Q8	Q9		
/5	/10	/10	/10	/10	/45	%

Section I

5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. Which expression is equal to $\int \tan x \, dx$?

(A) $\sec^2 x + C$

(B) $-\ln |\cos x| + C$

(C) $\frac{\tan^2 x}{2} + C$

(D) $\ln |\sec x + \tan x| + C$

2. Which of the following is an expression of $\int x^3 e^{x^4} \, dx$?

(A) $x^3 e^{x^4} + C$

(B) $e^{x^4} + C$

(C) $4e^{x^4} + C$

(D) $\frac{1}{4}e^{x^4} + C$

3. The diagram shows a cube of side 2 units.

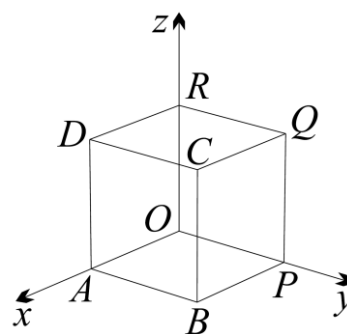
What is the length of the diagonal OC ?

(A) $\sqrt{6}$

(B) $2\sqrt{2}$

(C) $2\sqrt{3}$

(D) $3\sqrt{2}$



4. Which of the following is the first line of working for the expression:

$$\int x \cos x \, dx$$

(A) $-x \sin x - \int \cos x \, dx$

(B) $-x \cos x - \int \sin x \, dx$

(C) $x \sin x - \int \cos x \, dx$

(D) $x \cos x - \int \sin x \, dx$

5. Find any values of λ for which $\underline{a}$ and $\underline{b}$ are perpendicular if

$$\underline{a} = \begin{bmatrix} 1 + \lambda \\ 2\lambda \\ \lambda + 2 \end{bmatrix} \text{ and } \underline{b} = \begin{bmatrix} 2 \\ 5 - 4\lambda \\ 2\lambda - 4 \end{bmatrix}$$

(A) $\lambda = 0$

(B) $\lambda = -1$

(C) $\lambda = 1$

(D) λ has no solution

End of Section I

Section II

60 marks

Attempt Questions 6–9

Allow about 1 hour for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) Use a SEPARATE writing booklet.

- (a) The points P and Q have position vectors $1\hat{i} - 2\hat{j} - 3\hat{k}$ and $4\hat{i} - 5\hat{j} - 6\hat{k}$ respectively. Find

(i) $\overrightarrow{PQ}$ 1

(ii) $\overrightarrow{QP} - 2\overrightarrow{PQ}$ 1

(iii) $|\overrightarrow{PQ}|$ 1

- (b) Evaluate 2

$$\int_0^{\frac{\pi}{3}} \cos^3 \theta \, d\theta$$

- (c) Find the Cartesian equation of the line 2

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -4 \end{bmatrix}$$

- (d) Use the substitution $u = \sqrt{x}$, or otherwise, evaluate 3

$$\int_0^1 \frac{1}{(1+x)\sqrt{x}} \, dx$$

End of Question 6

Question 7 (10 marks) Use a SEPARATE writing booklet.

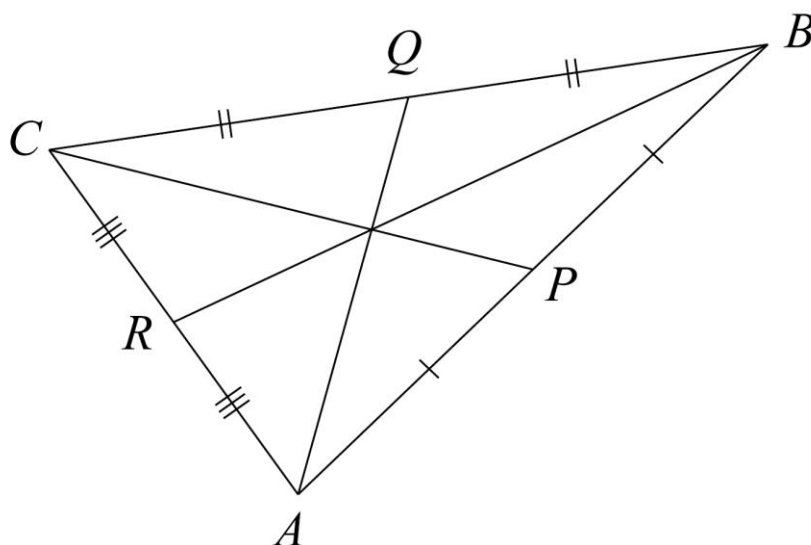
(a) Find

$$\int x \tan^{-1} x \, dx$$

3
[SL2]

(b) The diagram above shows $\triangle ABC$, where A , B and C have position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. The points P , Q and R bisect the intervals AB , BC and CA respectively.

3
[SL3]



(i) Show that $\overrightarrow{AQ} = \frac{1}{2}(\underline{c} + \underline{b}) - \underline{a}$ **1**

(ii) Hence show that $\overrightarrow{AQ} + \overrightarrow{BR} + \overrightarrow{CP} = 0$ **2**

(c) Let $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$, for intergeters $n \geq 0$

(i) Show that $I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}$ for $n \geq 2$ **2**

(ii) Hence evaluate $I_2 = \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$ **2**

End of Question 7

Question 8 (10 marks) Use a SEPARATE writing booklet.

(a) Determine $\int \sqrt{\frac{2-x}{2+x}} dx$ **3**

(b) Given $u = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} -7 \\ 6 \\ -5 \end{bmatrix}$

(i) What is the scalar product of the two vectors **1**

(ii) Hence find the projection of u onto v **2**

(c) Using the substitution $t = \tan \frac{\theta}{2}$ find **4**

$$\int_0^{\frac{\pi}{2}} \frac{1}{4 \sin \theta - 2 \cos \theta + 6} d\theta$$

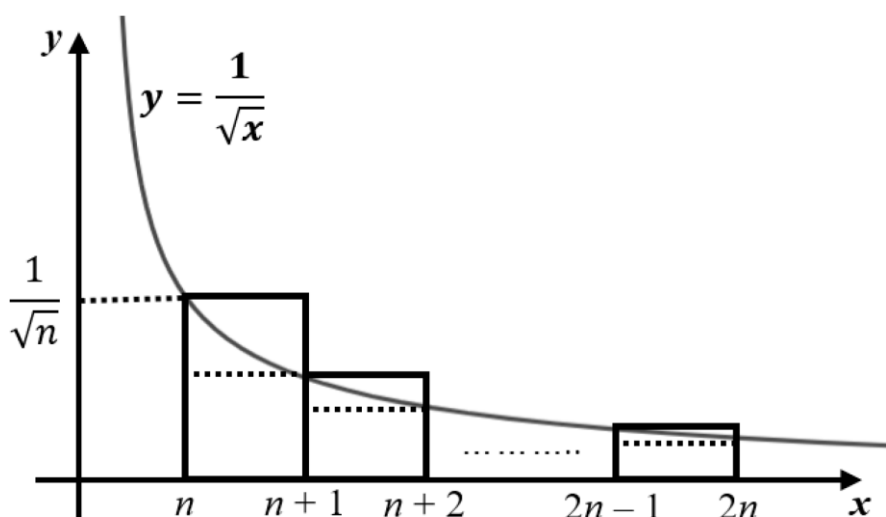
End of Question 8

Question 9 (10 marks) Use a SEPARATE writing booklet.

- (a) Find the exact values of x such that the angle between the vectors $\underline{a}$ and $\underline{b}$ is 60° if: 3

$$\underline{a} = \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} \text{ and } \underline{b} = \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix}$$

- (b) In the diagram below, the graph of $y = \frac{1}{\sqrt{x}}$ has been drawn, and n rectangles have been constructed between $x = n$ and $x = 2n$, each of width 1 unit



- i) Show that $\int_n^{2n} \frac{1}{\sqrt{x}} dx = 2\sqrt{n}(\sqrt{2} - 1)$ 1

- ii) Let $S_n = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n}}$, show that 4

$$2\sqrt{n}(\sqrt{2} - 1) + \frac{1-\sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

- iii) Hence find, correct to three decimal places

$$\frac{1}{\sqrt{1\,000\,000}} + \frac{1}{\sqrt{1\,000\,001}} + \frac{1}{\sqrt{1\,000\,002}} + \dots + \frac{1}{\sqrt{2\,000\,000}} \quad 2$$

End of Paper

Extra Writing Space

Please clearly indicate question number.

[illegible]

Multiple Choice Answer Sheet

Name _____

Colour the oval A, B, C or D that best answers the question.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↗

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

2021 EXT 2 AT3 SOLUTIONS

Q1 $\int \tan x \, dx = \int \frac{\sin x}{\cos x} dx = -\ln |\cos x| + C$ **B**

Q2 $\int x^3 e^{x^4} dx = \frac{1}{4} \int 4x^3 e^{x^4} dx = \frac{1}{4} e^{x^4} + C$ **D**

Q3 $|OC| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$ **C**

Q4 $u = x \quad v' = \cos x \quad \therefore u' = 1 \quad v = \sin x$
 $\therefore \int x \cos x \, dx = x \sin x - \int 1 \cos x \, dx$ **C**

Q5 $a = \begin{bmatrix} 1 + \lambda \\ 2\lambda \\ \lambda + 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 5 - 4\lambda \\ 2\lambda - 4 \end{bmatrix}$ and $a \perp b$
 $\therefore a \cdot b = 2 + 2\lambda + 10\lambda - 8\lambda^2 + 2\lambda^2 - 8 = 0$
 $\therefore \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2 = 0 \therefore \lambda = 1$ **C**

Q6a) i) $\overrightarrow{PQ} = \begin{bmatrix} 4 - 1 \\ -5 - (-2) \\ -6 - (-3) \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix}$
 ii) $\overrightarrow{QP} - 2\overrightarrow{PQ} = \begin{bmatrix} -3 \\ 3 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix} = \begin{bmatrix} -9 \\ 9 \\ 9 \end{bmatrix}$
 iii) $|\overrightarrow{PQ}| = \sqrt{3^2 + (-3)^2 + (-3)^2} = \sqrt{27}$

Q6b) $\int_0^{\frac{\pi}{3}} \cos^3 \theta \, d\theta = \int_0^{\frac{\pi}{3}} \cos^2 \theta \cos \theta \, d\theta$
 $= \int_0^{\frac{\pi}{3}} (1 - \sin^2 \theta) \cos \theta \, d\theta$
 $= \int_0^{\frac{\pi}{3}} \cos \theta - \sin^2 \theta \cos \theta \, d\theta$
 $= \sin \theta - \frac{1}{3} \sin^3 \theta \Big|_0^{\frac{\pi}{3}} = \frac{\sqrt{3}}{2} - \frac{1}{3} \times \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{3\sqrt{3}}{8}$

Q6c) $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -4 \end{bmatrix} \therefore x = 3 - \lambda \quad y = 5 - 4\lambda$
 $\therefore y = 5 - 4(3 - x) \therefore y = 4x - 7$

Q6d) $u = \sqrt{x} \therefore du = \frac{1}{2\sqrt{x}} dx$
 $\therefore \int_0^1 \frac{1}{(1+x)\sqrt{x}} dx = 2 \int \frac{1}{1+u^2} du = 2 \tan^{-1} u$
 $= 2 \tan^{-1} \sqrt{x} \Big|_0^1 = \frac{\pi}{2}$

Q7a) Let $u = \tan^{-1} x \quad v' = x$

$$\therefore u' = \frac{1}{1+x^2} \quad v = \frac{x^2}{2}$$

$$\therefore \int x \tan^{-1} x \, dx = \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \times \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{1+x^2-1}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int 1 - \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} [x - \tan^{-1} x] + C$$

Q7b) i) $\overrightarrow{AQ} = \overrightarrow{AC} + \frac{1}{2} \overrightarrow{CB} = (c - a) + \frac{1}{2}(b - c)$
 $= c - a + \frac{1}{2}b - \frac{1}{2}c = \frac{1}{2}c - a + \frac{1}{2}b$
 $= \frac{1}{2}(c + b) - a$

ii) As $\overrightarrow{AQ} = \frac{1}{2}(c + b) - a$

Then $\overrightarrow{BR} = \frac{1}{2}(a + c) - b$ and $\overrightarrow{CP} = \frac{1}{2}(a + b) - c$
 $\therefore \overrightarrow{AQ} + \overrightarrow{BR} + \overrightarrow{CP} =$

$$\frac{1}{2}(c + b) - a + \frac{1}{2}(a + c) - b + \frac{1}{2}(a + b) - c = 0$$

Q7c) i) $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$ Let $u = x^n; \quad v' = \sin x$
 $\therefore u' = nx^{n-1} \quad v = -\cos x$

$$\therefore I_n = [-x^n \cos x]_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx$$

Let $u = x^{n-1}; \quad v' = \cos x$

$$\therefore u' = (n-1)x^{n-2}; \quad v = \sin x$$

$$\therefore I_n = n \left([x^{n-1} \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1)x^{n-2} \sin x \, dx \right)$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

$$\text{Q7c) ii) } \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx = 2 \left(\frac{\pi}{2} \right) - 2 \times I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} \sin x \, dx = [-\cos x]_0^{\frac{\pi}{2}} = 1$$

$$\therefore \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx = \pi - 2$$

$$\text{Q8a) } \int \sqrt{\frac{2-x}{2+x}} \, dx \times \frac{\sqrt{2-x}}{\sqrt{2-x}} = \int \frac{2-x}{\sqrt{4-x^2}} \, dx$$

$$= \int \frac{2}{\sqrt{4-x^2}} \, dx - \int \frac{x}{\sqrt{4-x^2}} \, dx$$

$$= 2 \int \frac{1}{\sqrt{4-x^2}} \, dx + \frac{1}{2} \int \frac{-2x}{\sqrt{4-x^2}} \, dx$$

$$\text{Let } u = 4 - x^2 \quad du = -2x \, dx$$

$$\therefore \frac{1}{2} \int \frac{-2x}{\sqrt{4-x^2}} \, dx = \frac{1}{2} \int \frac{1}{\sqrt{u}} \, du = \frac{1}{2} [2\sqrt{u}]$$

$$\therefore \int \sqrt{\frac{2-x}{2+x}} \, dx = 2 \sin^{-1} \frac{x}{2} + \sqrt{4-x^2} + C$$

$$\text{Q8b) i) } \underline{u} \cdot \underline{v} = (3 \times -7) + (-2 \times 6) + (1 \times -5)$$

$$= -38$$

$$\text{Q8b) ii) } \text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{\underline{v} \cdot \underline{v}} \underline{v}$$

$$= \frac{-38}{(-7)^2 + 6^2 + (-5)^2} \underline{v}$$

$$= \frac{-19}{55} \underline{v}$$

$$\therefore \text{proj}_{\underline{v}} \underline{u} = \frac{-19}{55} \times \begin{bmatrix} -7 \\ 6 \\ -5 \end{bmatrix} = -\frac{133}{55} \begin{bmatrix} 7 \\ 6 \\ 5 \end{bmatrix}$$

$$\text{Q8c) } \int_0^{\frac{\pi}{2}} \frac{1}{4 \sin \theta - 2 \cos \theta + 6} \, d\theta$$

$$t = \tan \frac{\theta}{2} \quad \theta = 0, t = 0 \quad ; \quad \theta = \frac{\pi}{2}, t = 1$$

$$= \int_0^1 \frac{1}{\frac{8t}{1+t^2} - \frac{2(1-t^2)}{1+t^2} + 6} \times \frac{2}{1+t^2} \, dt$$

$$= \int_0^1 \frac{1}{4t-1+t^2+3+3t^2} \, dt = \int_0^1 \frac{1}{4t^2+4t+2} \, dt$$

$$= \int_0^1 \frac{1}{(4t^2+4t+1)+1} \, dt = \frac{1}{2} \int_0^1 \frac{2}{(2t+1)^2+1} \, dt$$

$$= \frac{1}{2} [\tan^{-1}(2t+1)]_0^1 = \frac{1}{2} (\tan^{-1}(3) - \frac{\pi}{4})$$

$$\text{Q9a) } \underline{a} \cdot \underline{b} = (1 \times x) + (2 \times 2) + (x \times 1) = 2x + 4$$

$$\underline{a} \cdot \underline{b} = \sqrt{1^2 + 2^2 + x^2} \sqrt{x^2 + 2^2 + 1^2} \times \cos 60^\circ$$

$$\therefore 2x + 4 = \sqrt{x^2 + 5} \times \sqrt{x^2 + 5} \times \frac{1}{2}$$

$$\therefore 4x + 8 = x^2 + 5 \quad \therefore x^2 - 4x - 3 = 0$$

$$\therefore x = \frac{4 \pm \sqrt{4^2 - 4 \times 1 \times (-3)}}{2} = \frac{4 \pm \sqrt{28}}{2} = 2 \pm \sqrt{7}$$

$$\begin{aligned} \text{Q9b)i) } \int_n^{2n} \frac{dx}{\sqrt{x}} &= [2\sqrt{x}]_n^{2n} \\ &= 2\sqrt{2n} - 2\sqrt{n} \\ &= 2\sqrt{n}(\sqrt{2} - 1) \end{aligned}$$

$$\text{Q9b) ii) } \text{Area of the lower rectangles} < \text{Area under the curve}$$

$$\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}} < \int_n^{2n} \frac{dx}{\sqrt{x}}$$

$$S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

$$\text{Area under the curve} < \text{Area of the upper rectangles}$$

$$\int_n^{2n} \frac{dx}{\sqrt{x}} < \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}}$$

$$2\sqrt{n}(\sqrt{2} - 1) < \frac{1}{\sqrt{n}} + \left(\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}} + \frac{1}{\sqrt{2n}} \right) - \frac{1}{\sqrt{2n}}$$

$$2\sqrt{n}(\sqrt{2} - 1) - \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{2n}} < S_n \quad 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n$$

$$\therefore 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

$$\text{Q9b) iii)}$$

$$\text{Let } n = 1\,000\,000$$

$$2\sqrt{1\,000\,000}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2\,000\,000}} < S_{1\,000\,000} < 2\sqrt{1\,000\,000}(\sqrt{2} - 1)$$

$$828.4268 < S_{1\,000\,000} < 828.4271$$

$$\therefore S_{1\,000\,000} = 828.427 \quad (\text{to 3 decimal places})$$

$$\frac{1}{\sqrt{1\,000\,000}} + \frac{1}{\sqrt{1\,000\,001}} + \dots + \frac{1}{\sqrt{2\,000\,000}} = \frac{1}{\sqrt{1\,000\,000}} + S_{1\,000\,000}$$

$$= 828.428 \quad (\text{to 3 decimal places})$$